

Prime Factorization & Exponent Methods

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The Intuition Prime Factor and Exponent Methods

One of the most fundamental rules of arithmetic guarantees that every nonzero positive integer (excluding 1) can be uniquely factored into a product of prime numbers and their powers.

This is very often the starting point of working on a math competition problem because you dig into the core components of elements you are working with to better understand them.

Formal Definitions and Notation

The definition of the **fundamental theorem of arithmetic** is:

Let $n \in \mathbb{Z}^+$ be an integer. Its prime factorization is uniquely given by:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} = \prod_{i=1}^k p_i^{e_i}$$

where $p_1 < p_2 < \cdots < p_k$ are prime numbers, and $e_i \geq 1$ are positive integer exponents.

Another basic function that you should know is one for counting the amount of divisors it has by choosing every possible combination of prime factors; for the integer defined as $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$:

Number of Divisors $d(n)$: The total count of positive factors is found by choosing an exponent from 0 to e_i for each prime:

$$d(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$$

As well as in similar manner

Sum of Divisors $\sigma(n)$: The sum of all positive factors expands into:

$$\sigma(n) = (1 + p_1 + \cdots + p_1^{e_1})(1 + p_2 + \cdots + p_2^{e_2}) \cdots (1 + p_k + \cdots + p_k^{e_k})$$

Other quick rules you should familiarize with are:

Perfect Squares: An integer n is a perfect square if and only if every prime exponent e_i in its prime factorization is even.

The same rule holds for cubes and so on but with their respective power divisibility.

GCD and LCM via Prime Factorization:

$$\gcd(a, b) = \prod p_i^{\min(e_i, f_i)}$$

$$\text{lcm}(a, b) = \prod p_i^{\max(e_i, f_i)}$$

Where $a = \prod p_i^{e_i}$ and $b = \prod p_i^{f_i}$

Scaffolded Examples

Direct Application (Easy)

Find the total number of positive divisors and the sum of positive divisors of $n = 180$.

Solution:

First, find prime factorization of 180 manually

$$180 = 18 \cdot 10 = (2 \cdot 3^2) \cdot (2 \cdot 5) = 2^2 \cdot 3^2 \cdot 5^1$$

Using the divisor counting formula:

$$d(n) = (e_1 + 1)(e_2 + 1)(e_3 + 1) = d(180) = (2 + 1)(2 + 1)(1 + 1) = 18$$

Using the sum of divisors formula:

$$\sigma(n) = (1 + 2 + 2^2)(1 + 3 + 3^2)(1 + 5) = 546$$

So our answers are:

18 and 546

respectively.

Slightly Harder Problem (Medium)

Find all positive integers n such that $2^n + 121$ is a perfect square.

Solution:

Suppose $2^n + 121 = m^2$ for some positive integer m .

Rearranging gives a difference of squares:

$$m^2 - 121 = 2^n \implies (m - 11)(m + 11) = 2^n$$

By unique prime factorization, any factor of a power of 2 must itself be a power of 2.

Let WLOG where $a < b$ and $a + b = n$:

$$m - 11 = 2^a \quad \text{and} \quad m + 11 = 2^b$$

Subtracting the two equations eliminates m :

$$(m + 11) - (m - 11) = 2^b - 2^a \implies 22 = 2^a(2^{b-a} - 1) = 2 \cdot 11$$

And since $b > a$, we know $2^{b-a} - 1$ must be an odd integer. Comparing the even power of 2 and odd factors on both sides:

$$2^a = 2 \implies 2^{b-1} - 1 = 11 \implies 2^{b-1} = 12$$

Since 12 is not a power of 2, there are no integer solutions for b . Thus, there are:

No solutions

Olympiad Application (Hard)

Find all pairs of positive integers (a, b) such that $a^2b + a + b$ is divisible by $ab^2 + b + 1$.

Solution:

We start by first establishing that b and $ab^2 + b + 1$ are coprime.

Suppose $d = \gcd(b, ab^2 + b + 1)$. Then $d \mid b$, which implies $d \mid b(ab + 1)$, so $d \mid (ab^2 + b)$.

Since d must also divide $ab^2 + b + 1$, it must divide their difference:

$$d \mid (ab^2 + b + 1) - (ab^2 + b) \implies d \mid 1$$

Thus, $\gcd(b, ab^2 + b + 1) = 1$.

Because b shares no common prime factors with the divisor $ab^2 + b + 1$, multiplying the numerator in the given valid fraction

$$ab^2 + b + 1 \mid a^2b + a + b$$

by b creates an equivalent divisibility relation without introducing extraneous solutions:

$$ab^2 + b + 1 \mid a^2b + a + b \iff ab^2 + b + 1 \mid b(a^2b + a + b)$$

Working out gives:

$$ab^2 + b + 1 \mid a(ab^2 + b + 1) + (b^2 - a) \implies ab^2 + b + 1 \mid b^2 - a$$

We now analyze the possible values of $b^2 - a$:

Case 1: $b^2 - a = 0$

$\implies a = b^2$; Testing $a = b^2$ in the original expression:

$$a^2b + a + b = b^5 + b^2 + b = b(b^4 + b + 1)$$

and

$$ab^2 + b + 1 = b^4 + b + 1$$

Since $b^4 + b + 1 \mid b(b^4 + b + 1)$ holds for all positive integers b , (b^2, b) is a valid solution family.

Case 2: $b^2 - a > 0$

For a positive integer to be divisible by another, the dividend must be at least as large as the divisor:

$$b^2 - a \geq ab^2 + b + 1$$

However, since $a \geq 1$ and $b \geq 1$, we have $ab^2 + b + 1 > b^2 > b^2 - a$, creating a contradiction.

Case 3: $b^2 - a < 0$

Let $a - b^2 > 0$ for the sake of contradiction. Divisibility requires:

$$a - b^2 \geq ab^2 + b + 1 \implies a(1 - b^2) \geq b^2 + b + 1$$

Since $b \geq 1$, we have $(1 - b^2) \leq 0$, while $b^2 + b + 1 > 0$.

A non-positive quantity cannot be greater than or equal to a positive quantity, so no solutions exist here.

Thus, the complete set of solutions is:

$$(b^2, b) \text{ for any strict positive integer } b$$

Reference Guide

Exponent manipulation and prime decomposition allow you to reduce high-degree polynomial expressions and complex divisibility statements down to fundamental properties of prime integer factors.

Important usages are Prime Factorization Identities:

-**Divisor Count** and **Divisor sum** as seen above.

What is also useful are Algebraic & Bounding Techniques:

-**Algebraic Factorization** like difference of perfect squares can reveal underlying structures formed by prime factors.

-**Coprime Component Splitting:** If $X \cdot Y = p^k$ and $\gcd(X, Y) = 1$, then one factor must be 1 and the other p^k .

-Size Bounds: Use divisibility size inequalities ($A \mid B \implies |A| \leq |B|$ for $B \neq 0$) to eliminate impossible exponent ranges.

Key Ideas

Simplify messy exponents and seemingly random numbers into their factors to see underlying structure

Practice Problems